

RobotX V1 Boundary Audit

A complete Phase 0–15 work inventory, an executed trace of the real core runtime, and one finite line drawn between a working engineering V1 and everything that is V2.

BRANCH / HEAD feature/dashboard · 09e91a5	SOURCE DIGEST d033038c... · 573 files	\$24 GATE TABLE 0 green · 17 red · 7 not eval.	V1 BLOCKING ITEMS 4 — 1 owner, 3 engineering
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01 Executive summary

The engine is not blocked by a missing subsystem. Every layer of the Tier 0 decision path is written, unit-tested and — where a database is involved — verified against live PostgreSQL. What is missing is **one leaf and one wire**: a `route(parts)` function, and the composition-root construction that would inject it. Everything else that is open is either evidence, calibration, certification, an external decision, or Tier 2.

THE SINGLE SENTENCE THAT DECIDES V1

A request can traverse the system as far as a durable `WorkQueue` row and then stops permanently, at `src/workers/leaderWorkers.js:151` — `UNCOMPOSABLE.coordinator`. Nothing else in the build reads that queue. The round loop, the solver, commitment, dispatch and supervision are all implemented and reachable *by injection*, and none of them has a production caller because the injection never happens.

Six findings that change the picture

1. **The correctness half of formal verification already passes at the capacity V1 actually ships.** `commitment_c1` and `lifecycle_c1` both close exhaustively with every declared property PASS. Capacity 2 and 3 exist to check *chaining* — Tier 2 mechanism T2-03, kill switch thrown, Phase 16e. **B-M's unconverged half is evidence for a mechanism V1 does not enable.**
2. **There is one genuine, unfixed correctness defect on the core decision path, and it is in no register.** Phase 9 handed limitation **L3** to Phase 10; Phase 10's closure never mentions it. `solve/round.js:248` still reads `expansion.achievedGapMilliCU ?? 0n` — coercing an *unproven* search gap to zero, which a decision record reads as "proven optimal". Phase 9 fixed the arithmetic precisely so this coercion would stop being necessary.
3. **B1 is not one item, and only part of it is on the V1 path.** Engine *selection* (Step 5, an ADR on benchmark evidence) is external and V2. A `route source` is a hard V1 runtime dependency. These are separable, and conflating them is what has made B1 look unbreakable.
4. **No configuration exists at all.** `service.defaultSnapshot()` returns `spatial: null`, `shards: null`, `bindings: {}`. Even with a router, there is no shard to enable and no region to bind, so the cutover gate refuses every request. This is D1, and it is an owner decision.
5. **The canonical documents describe a tree that is not checked in.** All five name HEAD `7335260`; HEAD is `09e91a5`, three commits later. The X7 fix they report as verified exists only as an **uncommitted** change to `formal/lifecycle.tla`.
6. **Two named modules were never built and appear in no Phase 15 register.** `src/engine/stores/` and `src/engine/deps/` are empty directories. `stores/roles.js` (Phase 3, invariant I16) and `deps/registry.js` + `circuitBreaker.js` (Phase 12) are named in `engine/ARCHITECTURE.md` and exist nowhere.

THE V1 ARITHMETIC

4 items block V1. One is an owner decision that cannot be delegated (what `route` talks to, and against which region). Three are engineering tasks Claude can complete, two of which are gated behind that decision. **Everything else — all 8 non-green §24 gates, B8's 39 findings, B-P, B-O, X3, A9, N16, N17, and every Phase 16 item — is V2.**

02 Phase 0–15 status matrix

Status here separates four things the documentation frequently merges: *phase closed*, *deliverable complete*, *downstream dependency open*, and *production certification open*. A phase can be closed and still have shipped a module that nothing calls.

PHASE	PURPOSE	STATUS	DELIVERABLE COMPLETE?	WHAT IS STILL OPEN, AND WHOSE	AFFECTS
0	Setup, tier/parameter/tenet gates	CLOSED, UNVERIFIED	Yes	Never independently verified — <code>PHASE_0_INDEPENDENT_VERIFICATION.md</code> does not exist and never did. Gates are exercised continuously by every later suite	No
1	Config, units, determinism	CLOSED	Yes	—	No
2	Domain model + schema	CLOSED	Yes	Closed "with documented environmental limitation". 28 migrations apply from empty	No
3	Commitment core (L3)	CLOSED	Yes	<code>src/engine/stores/roles.js</code> named and never built (I16 — cache must never hold the only copy). In no register	Latent
4	Dispatch + agent protocol (L2)	CLOSED	Yes	Its FAIL verdict was two weeks stale at closure; only the backoff item was real	No
5	Supervision + reconciliation	RE-CLOSED	Yes	Owens its own harness staleness: <code>phase5ExpirySemantics.js:334</code> and <code>:1101</code> assert a pre-T1-04 world (2 of 317 V-9 checks)	No
6	Feasibility gate (L4)	CLOSED	Yes	—	Reachable the round
7	Energy + payload models	CLOSED	Yes	—	Reachable the round
8	Cost function + plan builder (L4)	CLOSED, MODULE MISSING	No	<code>src/engine/routing/client.js</code> was never built. Phase 8's own module, blocked by N25/N26 and B1. A9	YES

Phase	Purpose	Status	Deliverable Complete?	What is still open, and whose	Affects
				(assertVersionInKey uncalled) is deferred to it	
9	Candidates + admissible bound	CLOSED, HANDOFF UNCLOSED	Mostly	L3 handed to Phase 10 and never executed. N16 (no per-candidate LB filter) and N17 (truncation by enumeration order) remain open by design	YES (L3)
10	Round loop + solve (L4→L3)	CLOSED, HANDOFF DROPPED	Mostly	Never picked up Phase 9's L3. Closure document does not mention <code>achievedGap</code> or <code>searchGap</code> anywhere	YES
11	Observability, decision records	CLOSED	Yes	Counterfactual gate exists; its corpus is behind the composition root	No
12	Failure, degraded modes, invariants	CLOSED, MODULES MISSING	No	<code>deps/registry.js</code> and <code>deps/circuitBreaker.js</code> named and never built. Directory is empty. In no register	Latent
13	Sharding, leadership, single writer	CLOSED	Yes	Sizing bound 1 (round wall-clock) is unmeasured — the round loop's to close (H6)	No
14	Security, governance, privacy	CLOSED	Yes	<code>surrogateKeys</code> written by no engine path (F2)	No
T1-04	§17.4 ladder, §17.5 starvation	CLOSED	Yes	4 named parts remain with stated reasons: exercise missions (Tier 2), rate limiting, saturation→\$20.5, rung 8 modality. None is a \$24 gate	No
15	Verification, gates, cutover	IMPL CLOSED · RELEASE BLOCKED	Yes (code)	8 of 24 blocking gates not green. None of the 8 is code. B1, B8, B-P, B-O, B-M, X3, A9 open	Only via B
16	Tier 2 enablement	NOT STARTED	n/a	Nine modules and one subsystem (§16 reliability) must be <i>built</i> inside it. Sole prerequisite: Phase 15	No

THE PATTERN WORTH NAMING

Three phases closed while leaving a module they themselves named unbuilt (8, 12) or a handoff unexecuted (9→10). None of those three items appears in `PHASE_15_BLOCKERS.md`, because Phase 15's registers are Phase-15-scoped by construction. **"0 repository-owned and actionable" has always meant "0 Phase 15 items".**

03 Complete remaining work inventory

Every meaningful open item found across the blocker register, the closure checklist §C, residual observations, `ARCHITECTURE.md` §10, phase-closure limitation tables, ADR open decisions, and the Phase 16 definition. Classification uses only the nine letters A–I you defined.

ID	PH.	SUBSYSTEM	Item & Current State	Source	Claude?	Core?
B1-route	8/15	routing	No <code>route(parts)</code> source exists. <code>cellPairCache.read()</code> returns "no router is available" when <code>source.route</code> is not a function. <code>routing/client.js</code> absent	<code>leaderWorkers:151;</code> <code>cellPairCache:195</code>	After owner decision	YES
B1-comp	15	server.js	Composition root does not construct <code>evaluateExact</code> , <code>pricedCandidateFor</code> , <code>hopsForSequence</code> . Coordinator refused at every promotion	B1 required work item 4	After B1-route	YES
B1-cfg / D1	—	config	No region, shard or binding exists. <code>defaultSnapshot()</code> → <code>spatial:null, shards:null, bindings:{}</code> (verified live this pass)	ARCHITECTURE §6.2	No	YES
L3	9→10	solve/round	Unproven search gap coerced to 0n , which a decision record reads as "proven optimal". Phase 9 supplies <code>achievedGapProven:false</code> and a <code>problems</code> entry; <code>round.js</code> discards both	P9 closure §9 L3; <code>round.js:248</code>	Yes, now	YES
B1-cut	15	coordinator	Coordinator does not read the per-shard cutover switch, so Rollback A does not stop a draining coordinator. Filed under B1 item 5, deliberately unwritten	V-10 finding V10-2	With B1-comp	Yes, once live
B1-sel	—	routing	Engine selection. Steps 1/3/4 (deploy, benchmark, record) then Step 5 (ADR). Needs D1+D3+D8	<code>routing:readiness</code> BLOCKED	No	No (V1 needs a source, not a selection)

ID	PH.	SUBSYSTEM	ITEM & CURRENT STATE	SOURCE	CLAUDE?	CORE?
D3	—	domain	No fleet mobility/speed model. Only <code>MobilityModel1</code> is a seed whose <code>speedModel1</code> is a note deferring to this decision	<code>routing:readiness</code>	No	Affects route quality
D8	—	routing	No extract identity, vintage, cadence, re-contraction budget, bbox or margin	<code>routing:readiness</code>	No	No
D1-esc	—	spatial	Governance vacuum. Approved JSSATE boundary $\approx 0.1022 \text{ km}^2$ vs $\approx 0.7373 \text{ km}^2$ H3 res-8 cell $\rightarrow$ zero cells $\rightarrow$ V-8 fires. Owner refused both over-assigning containment modes; no escalation authority exists	Handoff §1.8.3–1.8.5	No	Blocks D1
B8	—	calibration	39 blocking findings , 250 entries (52 DERIVED / 160 PROVISIONAL / 38 UNCALIBRATED), 54 Safety-class. Re-measured this pass	<code>gate:calibration</code> exit 1	No — §22.3 forbids it	No
B-P	15	gates	4 PRODUCTION gates NOT_EVALUATED: invariants, simulator fidelity, soak, shadow agreement. Shadow cannot even <i>start</i> its 14-day clock (B1)	<code>release:verdict</code>	No	No
B-0	15	gates	3 ORGANISATIONAL gates NOT_EVALUATED. 0 attestations filed	<code>release-evidence.json</code>	No	No
B-M	15	formal	Split in §11 below. 2 of 6 configurations close. <code>commitment_c1 PASS · c2 / c3 UNKNOWN · lifecycle_c1 PASS · c2 / c3 untreated</code>	BM run record §17	Partly (mechanics)	c1 half: already discharged
X3	15	lifecycle	No <code>TASK</code> timer producer; §4.2 has no transition table. Spec is FROZEN — must not be invented	<code>phase15CurrentTree</code> G1	Must not	No
A9	8	routing	<code>assertVersionInKey</code> implemented, tested, uncalled. Discharge point is <code>routing/client.js</code>	<code>chargerReachabilityCache:124</code>	With B1-route	No — <code>read()</code> carries an independent cross-check
N16	9	candidates	No per-candidate lower-bound filter. <code>evaluateOne</code> computes <code>bound</code> then calls <code>evaluateExact</code> unconditionally (verified this pass, <code>expansion.js:248–260</code>). Cell-level pruning does exist	ARCH gap 5 / N16	Yes	Performance, not correctness
N17	9	candidates	§9.4 "truncate by lower bound" is truncation by enumeration order. Carries a §6.6 ordering design question	ARCH gap 6 / N17	Partly	Quality, not admissibility
P3-roles	3	stores	<code>src/engine/stores/</code> is an empty directory . <code>roles.js</code> owns I16	ARCH gap 3; verified this pass	Yes	Latent
P12-deps	12	deps	<code>src/engine/deps/</code> is an empty directory . <code>registry.js</code> , <code>circuitBreaker.js</code> absent	ARCH gap 4; verified this pass	Yes	Latent
N13	—	workers	11 worker rows name a cadence parameter absent from the register (verified this pass). 5 of them run today. <code>calibration</code> and <code>counterfactual</code> fall back to hard-coded <code>300_000 / 86_400_000</code> ms	ARCH gap 9; measured	Yes	No — observability workers
OP-8	—	gates	§26 invariant-observation window is undefined anywhere. I13 consumes it; no phase creates it	ARCH gap 10	No (a derivation)	No
F4	15	gates	<code>locality</code> gate classified <code>EVIDENCE.SUITE</code> but the T9 cross-scale claim is not build-closable	ARCH gap 11	Yes	No
§20.5	T1-04	intake	Admission control returns ADMIT for all 30 purpose×SLA combinations . Four controls, four <i>different</i> missing decisions (§27 #13 quotas; "supply" undefined; shed level unmapped; "affected classes" undefined)	Blockers §T1-04	No	No — fails open by design today
Seed	—	prisma	<code>seed.js</code> exits 0 after a partial seed. 3 entries carry <code>changeClass: "OPERATIONAL"</code> ; enum has 7 members, none of them that (verified this pass). <code>main().catch(console.error)</code>	Residual obs. 8	Yes — but it is a §22.1 governance choice	No — fixture path
P15-F7a	15	docs	Nothing binds a runbook to the API it documents. Two partial compensating controls exist and are the most that should be built	Residual obs. 1	No (correctly)	No
CI	15	ci	CI does not run <code>gate:composition</code> and states no reason. CI is green on a tree where a blocking §24 gate is RED	Residual obs. 7	Yes	No

ID	PH.	SUBSYSTEM	ITEM & CURRENT STATE	SOURCE	CLAUDE?	CORE?
V-5	15	evidence	<code>release-evidence.json</code> aged out — 17 records ~281 000 s old against an 86 400 s bound (re-measured this pass). Re-collection is the release owner's	<code>release:verdict</code>	Mechanically yes	No
16a-16i	16	Tier 2	Nine sub-phases. Nine modules plus \$16 reliability must be built , not switched. 3 mechanisms have no sub-phase at all (OD-2)	Plan §3 Phase 16	Not now	No
OD-2..5	16	governance	OD-2 (3 ungated Tier 2 mechanisms), OD-3 (§20.1 budget composition), OD-4 (B10 Forecast Service — UNASSIGNED), OD-5 (B7 MIP solver)	Plan §6.4	No	No

04 Actual core runtime path

Traced against the repository, not against the plan. Each link states whether it is implemented, whether it is *composed* (the distinction that matters here), and whether a real request reaches it today.

01	HTTP intake — <code>POST /api/tasks/assign</code> <code>tasks.routes</code> → <code>tasks.controller.assignTask:83</code> → <code>task.service.assignTask</code> . Implemented, composed, reachable, tested.
02	Cutover gate — before any write <code>cutoverEnabled.describe({snapshot, shard})</code> . Refuses with <code>503 ENGINE_NOT_LIVE</code> <i>before</i> the row is created, deliberately. Today it always refuses — not because of a code defect, but because no snapshot contains a region, a shard or a binding.
—	Everything below this line is proven by test and live-DB harness, and is unreachable in production today. The steps are ordered as they would execute with the gate open.
03	Task admission — <code>admitToRound</code> Mission upsert (idempotent on <code>Task.taskId</code>), then <i>one transaction</i> : Leg upsert + <code>superviseQueuedEntry</code> arming the §4.5 <code>QUEUED</code> deadline and writing <code>Leg.slaDeadline</code> . T1-04's load-bearing find. Verified 72/72 on live PostgreSQL.
04	§20.5 admission control — <code>intake.admit</code> → <code>admission.assess()</code> Complete and correct, and inert : <code>admissionInputs</code> is <code>{}</code> because no controller sets it, so <code>assess()</code> returns ADMIT for all 30 purpose×SLA combinations with <code>shedLevel</code> pinned at NOMINAL. Fails open by design; four separate undecided semantics behind it.
05	Durable <code>WorkQueue</code> row "The request path's last act is a durable <code>WorkQueue</code> row, and the round path's first act is to read it." The first half executes.
06	HALT — the round path's first act never happens <code>leaderWorkers.COMPOSERS.coordinator()</code> returns <code>{ok:false, ...UNCOMPOSABLE.coordinator}</code> at every leadership promotion. <code>gate:composition</code> reports it as the single violation across 19 workers, owner EXTERNAL . Nothing else in the build reads <code>WorkQueue</code> . A Leg admitted today stays <code>QUEUED</code> until its deadline fires, escalates up §17.4's ladder, re-arms, and eventually exhausts to FAILED.
07	Round L4 — <code>solve/round.plan(deps, input)</code> Written, tested, deterministic. Consumes <code>deps.expandCandidates</code> , <code>deps.pricedCandidateFor</code> , <code>deps.planState</code> , <code>deps.budgets</code> by injection. Carries defect L3 at line 248.
08	Candidate expansion — <code>candidates/expansion.js</code> §6.3 seven-tier search. H3 geometry is pure (<code>h3-js</code>) and needs no published map. Search-gap arithmetic was corrected by Phase 9 F9-4 and is conservative. N16/N17 remain: no per-candidate LB filter, truncation by enumeration order.
09	Feasibility → cost → plan builder <code>evaluateExact</code> → Phase 6 gate → Phase 8 plan builder and $\emptyset$. All implemented and closed.
10	ROOT CAUSE — the routing leaf <code>plan/insertion.js</code> → <code>planBuilder.hopsForSequence</code> → <code>routing/cellPairCache.hopsFor</code> → <code>source.route(parts)</code> . When <code>source.route</code> is not a function, <code>read()</code> returns "no router is available and the entry is not cached" . No production route exists anywhere. <code>src/engine/routing/</code> holds three cache modules and no client. <code>services/routing.service.js</code> is the legacy DTARO A*/Mapbox path — a different contract, not this seam.
11	Solve → commitment (L3) min-cost flow / cost scaling, then Phase 3's <code>commit()</code> . Commitment is verified against live PostgreSQL and exhaustively model-checked at capacity 1.

12	Dispatch, supervision, reconciliation <code>outbox</code> , <code>timer</code> , <code>reconciler</code> — all three compose and run today on leadership. The timer worker drives §17.4's escalation ladder. They supervise a queue that never drains.
13	Persistence and observability 28 migrations apply from empty; 74 domain tables + <code>LadderEscalation</code> . Decision records, SLIs, metrics all implemented.

THE ANSWER TO "EXACTLY WHERE DOES IT STOP"

Two places, and only two. (1) `cutoverEnabled.describe` refuses because no configuration exists — a *data* gap. (2) `leaderWorkers.js:151` refuses because `source.route` does not exist — a *module* gap. Every other link in the chain is built, tested and closed. There is no third gap.

05 Currently working core functionality

Measured on this tree, this pass, unless marked. This is what you already own.

CAPABILITY	STATE	EVIDENCE
Host platform — robot auth, telemetry ingest, dashboard feed, obstacle handling, rerouting, simulator, operator API	LIVE	README "Everything else runs"
Legacy DTARO decision path	DELETED	<code>gate:legacy</code> PASS — 4 modules absent across 346 files
Composition root — <code>server.js</code>	EXISTS	19 workers registered · 9 SCHEDULED + 3 of 4 LEADER_ONLY start = running (verified this pass)
Cutover authority — 10 modules, one-directional, register-resolved bounds	CLOSED	80/80 live-DB across 4 harnesses
§24 gate table — 24 gates, all blocking, correct algebra	IMPLEMENTED	<code>release:verdict</code> renders 24 and detects its own staleness
Request → queue path with a supervised deadline	VERIFIED	<code>verify:t104</code> 72/72 live PostgreSQL
§17.4 escalation ladder + operator capacity + §17.5 starvation	COMPOSED	<code>leaderWorkers.timer</code> → <code>expiryActions</code> → <code>ESCALATION_LADDER</code> ; 9
Commitment core — fences, idempotency, guards	EXHAUSTIVE	<code>commitment_c1</code> : 17 991 520 states / 2 375 660 distinct / diameter 2 error
Leg lifecycle — safety, terminal-finality, liveness	EXHAUSTIVE	<code>lifecycle_c1</code> : 777 942 / 187 289 / depth 43 / 0 on queue · all thre conjuncts PASS UNCOMMITTED
Rollback + cutover runbooks, traced against the live API	VERIFIED	V-10: 5 defects fixed, 15/15 live-DB, 3 mutants killed
Phase 0–14 contract compatibility after the Phase 15 tree movement	VERIFIED	V-9: 7 harnesses, 315/317, both failures stale FINDING assertions
Migration chain from empty	GREEN	28/28, 0 failed, 0 rolled back
Full test suite	GREEN	162 suites / 7 275 tests / 0 failures (<i>documented 2026-08-30; not re-read-only pass</i>)
Build gates	7 of 8	<code>gate:composition</code> FAIL — 1 violation across 19 workers (re-run this

06 V1 must have

THE DEFINITION THIS AUDIT DERIVES

V1 = a working core assignment engine that executes its Tier 0 / Tier 1 workflow correctly and reproducibly at capacity 1, against one declared operating region, with every known limitation written down. Not "every §24 gate green". Not "Phase 15 release". Capacity 1 is not a compromise — chaining is Tier 2 (T2-03), its kill switch is thrown, and Phase 16e owns it.

Four items, and no others, meet the Part 7 test.

#	ITEM	WHY IT IS V1 MUST	WHO
V1-1	Decide what <code>route(parts)</code> talks to, and against which region — one recorded decision, in an ADR	Without it the core path has no leaf and the coordinator cannot be composed. <i>Failure means the engine cannot work at all.</i> This is not B1	OWNER

#	ITEM	WHY IT IS V1 MUST	WHO
		Step 5: V1 needs a declared, documented traversal <i>source</i> , not a benchmarked production selection	
V1-2	Build the routing client and the composition-root construction — <code>src/engine/routing/client.js</code> (§5.2 degradation ladder, §18.3 B6 uniform treatment, the <code>route(parts)</code> seam) and the <code>server.js</code> injection of <code>evaluateExact</code> , <code>pricedCandidateFor</code> , <code>hopsForSequence</code> , plus the coordinator's cutover-switch guard (B1 item 5) and the <code>assertVersionInKey</code> call site (A9)	The required core dependency is missing; the core path cannot be executed end to end. Note this discharges A9 and B1 item 5 as a side effect — both are deferred today only because this module does not exist	Claude 1
V1-3	Publish a configuration version containing a region, a shard and the cutover binding	Known missing core dependency. <code>defaultSnapshot()</code> returns nulls, so the cutover gate refuses every request regardless of the router. Do not use <code>SEED_SPATIAL_MAP</code> / <code>RGN-BLR</code> — it is a Phase 2 containment demonstration	OWNED decide publish
V1-4	Fix L3 — <code>solve/round.js:248</code> . Propagate <code>achievedGapMilliCU: null</code> and <code>achievedGapProven: false</code> instead of coercing to <code>0n</code>	Known correctness defect on the core path. Every decision record the engine produces carries this number and §21.4 aggregates it into an SLI. Coercing an unproven bound to zero writes "proven optimal" into the record — the exact false guarantee §6.4 calls worse than no bound. Its caller already exists , so no seam is manufactured	Claude

V1-1 IS THE ONLY GENUINE FORK IN THIS PROJECT

Everything else follows from it. The repository is deliberately built to refuse a fabricated router, and that refusal is correct — *"starting the coordinator against an invented router would assign real work on invented travel times."* But that refusal is a guard against **Claude** inventing one. It is not a guard against **you** declaring one, in an ADR, with its limitations stated. Those are different acts, and the second is available to you today.

07 V1 verify

Already implemented; must be re-checked before the V1 freeze because V1-2 changes the composition root and V1-4 changes the round.

#	CHECK	PASS CONDITION
VV-1	<code>npm test</code>	exit 0, 0 failures, 0 skips
VV-2	<code>npm run gates</code>	exit 0 — all 8. <code>gate:composition</code> turns green when the coordinator composes; that is the V1 signal
VV-3	One real request, end to end, against a disposable PostgreSQL cluster	<code>POST /api/tasks/assign</code> → Leg QUEUED → coordinator round claims it → candidates expanded → commit → dispatch. A durable Commitment row exists and its decision record carries a proven gap. This harness does not exist yet and is part of V1-2's verification
VV-4	Six live-DB harnesses (167/167) re-run	167/167 on a disposable cluster, port ≠ 5432, never Neon
VV-5	V-9 re-run — the 4 harnesses whose <code>require()</code> closure touches changed files	No regression. V1-2 and V1-4 change <code>round.js</code> , <code>server.js</code> and add a module; re-derive the scope by measurement, do not assert it
VV-6	Mutation testing per implementation pass (V-6)	Every mutant built for V1-2 and V1-4 is killed. The L3 mutant is the obvious one: restore the <code>?? 0n</code> and a test must fail
VV-7	Commit <code>formal/lifecycle.tla</code> and re-run <code>lifecycle_c1</code> as checked in	777 942 / 187 289 / depth 43 / 0 on queue, exit 0. The verified state currently exists only in an uncommitted working tree
VV-8	Source digest re-measured and recorded at the freeze commit	A single digest that the V1 freeze document quotes

08 V2 — external / operations

Class `E`. Not completable through repository work at any effort level.

ITEM	WHAT IS ACTUALLY MISSING	WHY NOT V1
B1 Steps 1/3/4/5	Deploy each candidate against a real region extract, build per-profile contraction hierarchies, benchmark on representative hardware, record, then choose in an ADR	V1 needs a <i>route source</i> , not a <i>procurement decision</i> . A bencil comparing engines is a production-quality question
D3	The agent classes operated, and per distinct mobility model §2.2's six elements with a real speed model over <code>roadClass</code> , <code>gradient</code> , <code>surface</code> , <code>payloadMass</code> , <code>congestion</code> , <code>weather</code>	Affects route <i>quality</i> , not whether the path executes. V1 runs declared model with stated limitations

ITEM	WHAT IS ACTUALLY MISSING	WHY NOT V1
D8	Extract identity, source, ISO vintage, refresh cadence, re-contraction downtime budget, <code>extract.bbox</code> , <code>extract.marginDegrees</code>	Pure operations metadata. Nothing in the decision path read
B8	39 Safety-class parameters derived from a stated accounting or measured basis, by a named calibration owner. Each finding names what it awaits — certifications, measured distributions, site-authority decisions	Calibration incompleteness with a working core runtime is excluded by your Part 7 rule. §22.3 forbids any automated p from changing a Safety-class parameter
RNSIT snapshot	The RNSIT boundary snapshot artefact and every D1 governance record — no approval, no Commercial confirmation, no Architecture approval exists	Governance, not runtime

09 V2 — specification / governance

Class **F**. A decision, not a commit. The specification is **FROZEN**; none of these may be answered by writing code.

ID	DECISION REQUIRED	BLOCKS
X3	A §4.2 <code>TASK</code> transition table, or an ADR declaring that <code>TASK</code> has no timed transitions	Closure item I-12 only, runtime behaviour
D1-esc	The coverage-semantics escalation has no target. Approved boundaries are smaller than one H3 res-8 cell, V-8 fires, <code>cardinalityException</code> cannot rescue it, both over-assigning containment modes are refused, and RobotX has no Architecture or approval authority. The escalation has no decision authority and none may be invented	D1 → B1 Steps 1/3/4/5
§20.5-a	What quantity <i>intake</i> measures as "supply". <code>feasibleSupply</code> has no producer anywhere; §3.4 forbids consulting the routing provider on the request path; the Availability Index is cache-tier and §18.5 rule 3 forbids promoting it	Global admission contr
§20.5-b	What "the affected classes" denotes in §17.4 — the phrase occurs once in 5 941 lines and is never defined — and whether the consequence is a shed verdict or a global decline	Saturation → admissor
§20.5-c	A published mapping from overload severity to shed rungs 1/2/3. §18.5 makes Shed Load binary; B19's entry condition has no registered threshold	Class-based shedding
§27 #13	Multi-tenancy model — drives quota design. <code>tenantId</code> is null on every request	Per-tenant quotas
Seed-gov	Add <code>OPERATIONAL</code> to <code>ConfigChangeClass</code> , or reclassify the 3 entries. A §22.1 register governance choice, not a mechanical one	<code>phase12LiveDataBase</code> empty cluster
OD-2	Enabling gates for churn pricing, post-solve local search, the duty-cycle regulariser — three Tier 2 mechanisms with a kill switch each and no ladder row, sub-phase or gate. Escalated at Phase 0; never answered	Completeness of Phase
OD-3	§20.1 budget composition — the frozen architecture states the statistic and the scope and no aggregation rule	Phase 16b; <code>scale_tar</code> having a defined target
Question A	Whether <code>STRANDED_*</code> is cancellable at all. Recorded during the X7 pass, deliberately unresolved, non-blocking	Nothing

10 V2 — production / certification

Class **D**. Required for a real production release. Not required for a working engineering V1, and your Part 7 rule excludes every one of them by name.

GATE / ITEM	STATE	WHAT DISCHARGES IT
<code>invariants_enforced</code> §26	NOT_EVALUATED	An observation window that does not exist anywhere in the repository (OP-8) pl operation
<code>simulator_fidelity</code> §24.4	NOT_EVALUATED — 7 models NOT_MEASURED, 6 safety-relevant (re-run this pass)	A one-sided study against realised production data. Do not synthesise one
<code>soak</code> §24.6	NOT_EVALUATED	A 72 h soak — and <code>release.soak_duration</code> is itself still PROVISIONAL (B8)
<code>shadow_agreement</code> §21.6	NOT_EVALUATED — the clock cannot start	14 days of live traffic. Gated by B1: the shadow worker cannot compose either. It releases this and it then takes 14 days
<code>safety_case_assembled</code> §24.7	NOT_EVALUATED	A filed attestation. The assembler exits 0 and files no evidence — an assembling case is not a discharged gate
<code>rollback_rehearsed</code>	NOT_EVALUATED	A rehearsal per <code>rollback.md</code> §5 in a declared non-production environment (AC REHEARSAL), with all six step keys

GATE / ITEM	STATE	WHAT DISCHARGES IT
V-5 — evidence re-collection	RED, aged out (~281 000 s vs 86 400 s bound)	The release owner running <code>release:gates</code> at a quiescent, committed tree. Rer 16/1/7 and still reads BLOCKED
F4 — <code>locality</code> gate class	Open	The T9 cross-scale claim is not build-closable; the gate's classification must ch before a production cutover
B-M items 4, 8, 10	Unsatisfied	A named human operator, boundedness acceptance, release-owner acceptance

11 V2 — future phase / Tier 2, and the B-M split

B-M, split as you asked — and the split is what moves the V1 boundary

COMPONENT	STATE	V1 OR V2, AND WHY
Model correctness	SETTLED — four findings, four transcription defects, zero defects in the shipped engine	B Already discharged. X4, X5, X6, X7 all closed; each time th module was the one that was wrong
Transcription defects X4 / X5 / X6 / X7	ALL CLOSED	B Commit them. X7's fix is uncommitted
<code>lifecycle_c1</code> evidence	PASS — 777 942 / 187 289 / depth 43 / 0 queued · Safety, TerminalIsFinal, Liveness (all three conjuncts)	B This is V1's formal evidence. Capacity 1 is the capacity V1
<code>lifecycle_c2</code> / <code>c3</code>	Untreated / non-convergent. <code>ladder</code> multiplies the space by up to <code>(LadderSteps+1)^(Legs)</code> ; the latch adds <code>2^(Legs)</code>	G Capacity 2/3 checks chaining — T2-03, kill switch thrown, 16e's own gate names "model check green at 2 and 3"
<code>commitment_c1</code> evidence	PASS — 17 991 520 / 2 375 660 / diameter 21, exhaustive	B V1's commitment evidence
<code>commitment_c2</code> / <code>c3</code>	UNKNOWN — did not converge (126 M and 21 M on queue; 23.6 GiB disk queue still growing on c2)	G Same tier argument, plus a genuine compute requirement
Tool provenance (items 1–3)	RECORDED — TLA+ 1.8.0, SHA-256 <code>eabd140a..533a</code> , JDK 20.0.2, LENOVO 21DJ i5-1235U, contended, machine slept during c2	B Already satisfied for the runs that closed
Raw evidence retention	PROVISIONAL — retained outside the repository; the repository prescribes no evidence directory and no file format	D The release owner's to confirm or redirect
Boundedness acceptance (item 8)	NOT ACCEPTED — <code>MaxTicks = 3</code> is a <i>global</i> budget, not per-Leg, and does not scale with Capacity while Legs does	B at capacity 1 only. Two Legs, three global ticks — you must whether that is still the system. This is the one B-M item V1 ge inherits
Named operator (item 4)	UNSATISFIED	D §24.7 attributability, for a production release
Property coverage — G5 gap	Guard G5 (cancellation, purpose-conditioned) has no counterpart in either model. Vacuous in both. No TLC run covers it	B Its evidence is <code>commitmentGuards.test.js</code> plus the live- Record the gap in the V1 freeze; do not close it with a model
Final acceptance (item 10)	NOT GIVEN	D
Release-gate integration	Structurally impossible today — a <code>java -jar tla2tools.jar</code> run cannot be filed against an <code>EVIDENCE.SUITE</code> gate (<code>COMMAND_MISMATCH</code>), and <code>formal/</code> is outside the digest scope	D Recorded, not to be fixed. Do not modify <code>gates.js</code> , <code>evi</code> or <code>sourceDigest.js</code> to accommodate a TLC run

Tier map

TIER	CONTENT	V1?
Tier 0 — safety core	Commitment, custody, lifecycle, feasibility, admissible bound, determinism	V1 CORE
Tier 1 — operational integrity	Supervision, reconciliation, dispatch, sharding, observability, §17.4/§17.5 fairness (T1-04), invariants	V1 CORE — all shipped and compos
Tier 2 — allocation quality	T2-01...T2-13: batch solving, multi-Leg columns, chaining, deferral, preemption, opportunity cost, churn pricing, local search, reliability pricing, duty cycle, repositioning, consolidation, cross-region	V2 — <code>every switch stays thrown</code>
T1-04 remainder	Exercise missions (needs Tier 2 §17.3), rate limiting into the human queue, saturation→§20.5, rung 8 modality	V2 — each has a stated reason, nor gate
Phase 16 enablement	Nine modules plus §16 reliability must be <i>built</i> , not switched. Three mechanisms have no sub-phase	V2 — prerequisite is Phase 15, and is not V1

PHASE 16 PREREQUISITE MAP

Every Phase 16 sub-phase has exactly one prerequisite: **Phase 15 complete**, which means the §24 table green — which means B1, B8, B-P and B-O. **V1 requires no Phase 16 item and enables no Phase 16 prerequisite**. 16a additionally needs B10 (Forecast Service, owner UNASSIGNED) or its `λ_zone` resolves permanently to the static priors it exists to replace; 16d needs B7 (column-regime MIP solver); 16e needs the capacity 2/3 model checks. None is manufactured here.

12 Residual non-blocking observations

Class **I**. Each was asked: *does this demonstrably break the V1 core runtime?* Every answer below is no, and none was promoted for sounding serious.

ITEM	VERIFIED STATE	WHY IT IS NOT V1
Seed partial failure	CONFIRMED LIVE this pass — 3 entries carry <code>changeClass:"OPERATIONAL"</code> , enum has HOT/TUNED/POLICY/SAFETY/STRUCTURAL/DERIVED/CONTRACTUAL. <code>main().catch(console.error)</code> at <code>seed.js:393</code> , exits 0 after 165 of 250 entries	Development/verification fixture path Moves no §24 gate. The fix is a §22.1 governance choice. Do not fabricate tl Mission row
<code>lifecycleModel.js</code> divergence	<code>:350</code> exempts <code>CANCEL_REQUEST</code> from <code>cancellationGuard</code> , so the executable checker admits a repeat the shipped engine refuses	Permissive in the checker, not the en ; <code>cancellation.js:119-130</code> writes th under a version-conditional <code>updateMa transitions.js:664-669</code> reads it; n <code>Backend/src/</code> clears it. Record it in t freeze
G-1 — no §4.2 TASK state written	Confirmed by <code>phase15CurrentTree</code> as a finding assertion	X3. Specification-owned, must not be
G-2 — coordinator and shadow uncomposable	Confirmed	This is V1-2, promoted to V1 MUST. No
G-3 — calibration gate and publish validator agree B8 blocks cutover	190 launch-gate findings; defaults refused by V9	B8. External
H-1 / M-1 and pass-2's X-C1 / X-C2	Carried forward, none permissive	Recorded in observation 6; no permiss direction
Runbook↔API binding (P15-F7a)	Structural. V-10 is its second demonstration — five drifts in the second runbook	Neither remedy should be built: digest <code>docs/</code> voids a 25-min collection on e edit; a prose-parsing gate is a future fe The two compensating controls are th stopping point
N13 cadence parameters	MEASURED THIS PASS — 11 workers name a parameter not among the 250 register entries. 5 run today . <code>calibration</code> uses <code>SCORE_INTERVAL_MS = 300_000</code> ; <code>counterfactual</code> uses <code>RUN_INTERVAL_MS = 86_400_000</code> . <code>reconciler</code> and <code>timer</code> have declared, reasoned, conservative fallbacks	Observability and calibration cadence: decision path. <code>ARCHITECTURE.md</code> 's "ir is now false for 5 rows — worth correc worth blocking V1
<code>app.locals.releaseEvidence</code> has no producer	Read as <code> {}</code> at <code>health.controller.js:319</code>	Fails closed — every gate reads as not
<code>gates.blockers()</code> ignores <code>unknownEvidence</code>	Confirmed	<code>evaluate()</code> folds it into <code>ok</code> and <code>ver</code> prints it
Evidence schema cannot distinguish failed from never-ran	Discovered when a gate process failed to start	Detected in practice; the affected colle voided, not published
CI omits <code>gate:composition</code>	7 of 8 steps, 6 on a push. No reason recorded	CI is not the §24 authority. But this is fixing at the V1 freeze , because after gate is the one that proves V1 works
Historical provenance — the 2026-08-15 TLC run	v1.8.0 was published 2026-08-21, six days <i>after</i> the run is dated. Offsetting: the 2026-08-31 <code>commitment_c1</code> run reproduced its state count, distinct count and diameter exactly	Historical record. Do not edit it . Whet counts toward the six is the release ov acceptance decision, and V1 does not <code>commitment_c1</code> was re-run
Documentation drift	See §14	Stale counts do not break a runtime

13 Owner actions

Things Claude cannot legitimately complete. A generated document is not real-world evidence, and none of these is satisfied by producing one.

ACTION	WHY CLAUDE CANNOT DO IT	EVIDENCE YOU WOULD NEED	BLOCKS V1?
OA-1 · Decide V1's traversal source — what <code>route(parts)</code> talks to, and against which region, recorded in an ADR with its limitations stated	The repository refuses to select, rank, recommend or hint at an engine — <code>b1Readiness.js</code> refuses in code, and so must I. A declaration is an act of authority, not of engineering	An ADR naming the source, its scope, what it is <i>not</i> (a benchmarked production selection), and the limitation V1 ships with	YES — the
OA-2 · Declare the V1 operating region and accept its cell-cover consequence	Inventing a region, boundary, CRS or kind is forbidden. The V-8 cell-cover problem has no escalation authority and none may be invented	A region with <code>regionId</code> , name, kind, CRS, version+date, an adopted boundary, and a stated decision on the sub-cell coverage question	YES
OA-3 · Accept the capacity-1 boundedness — <code>MaxTicks = 3</code> as a global budget across 2 Legs	§7.3a item 8 requires this to be signed, not assumed. "Whether a three-tick lifecycle is still the system" is a safety judgement	One written acceptance in the V1 freeze document, or a rejection with a stated alternative bound	YES — one
OA-4 · Name a calibration owner and derive 39 Safety-class values	§22.3 forbids any automated process from changing a Safety-class parameter	Per-parameter derivation from a stated accounting or measured basis	No — V2
OA-5 · Operate a fleet	Observation cannot be simulated. P15-E1 exists because malformed windows were being ADMITTED and PASSED on exactly these gates	A real soak, a real 14-day shadow window, a real invariant window, a real fidelity study	No — V2
OA-6 · File three attestations as a named human	Filing on anyone's behalf is forbidden, and <code>verdict.js</code> forces a gate RED when a build record and an attestation collide	Safety case, calibration, rollback rehearsal — each admitted by <code>evidence.admit()</code>	No — V2
OA-7 · Rehearse the rollback in a declared non-production environment	A rehearsal is an event, not a document. All six <code>REHEARSAL_STEPS</code> keys must be present or the record is refused	A record naming cutover, <code>automatic_rollback</code> , <code>no_decision_path_confirmed</code> , <code>artefact_rollback</code> , <code>recutover</code> , recorded — plus <code>recordedBy</code> / <code>approvedBy</code>	No — V2
OA-8 · Provision compute for <code>commitment_c2</code> / <code>c3</code>	This workstation could not converge them; 23.6 GiB of disk queue was still growing	Machine time and a machine statement	No — V2 (Tier capacity)
OA-9 · Answer X3, OD-2, OD-3, §20.5's four semantics, the seed governance question	The specification is FROZEN. Writing an answer would be writing specification	A recorded decision or an ADR per item	No — V2

ONLY THREE OWNER ACTIONS BLOCK V1

OA-1, OA-2, OA-3. Two are decisions you can record today; the third is one sentence of acceptance. Everything else on this list can wait for V2 without making V1 dishonest, provided V1 says plainly that it is an engineering release and not a certified production one.

14 Documentation contradictions and stale information

Current documentation defects — correct these

#	WHERE	WHAT IT SAYS	WHAT IS TRUE
DD-1	All five canonical Phase 15 docs, identity rows	HEAD 7335260	HEAD is 09e91a5 — three commits later (ef0d65f , 22411e8 , 09e91a5) digest d033038c... /573 is still correct (re-measured this pass), which is wr noticed
DD-2	Working tree vs. record	X7 "fixed in formal/lifecycle.tla and verified"	The fix is uncommitted. git status shows formal/lifecycle.tla and formal/README.md modified. The verified lifecycle_c1 PASS describes that is not checked in
DD-3	PHASE_15_MASTER.md §7 blocker table, X6 row	"X6 — NEW, OPEN ... deliberately not decided in the pass that found it"	X6 was CLOSED on 2026-08-31 — as the same document's own banner sa lines above. A contradiction inside the canonical set, whose §7.1 asserts the five canonical documents: none known"
DD-4	PHASE_15_CLOSURE_CHECKLIST.md , final verdict paragraph	"X3 and X6 require specification or formal-verification decisions"	X6 is closed. Only X3 remains
DD-5	PHASE_15_BLOCKERS.md B-M results table	Lifecycle rows record "FAIL — Liveness violated (X6)"	Superseded twice since. lifecycle_c1 PASSES; the strike-through conve not extended to these rows
DD-6	ARCHITECTURE.md §10 gap 16	" fairness/ladder.js , operatorCapacity.js and agentStarvation.js do not exist ... still blocking"	All three exist and are composed. T1-04 closed 2026-08-30 (verified this

#	WHERE	WHAT IT SAYS	WHAT IS TRUE
DD-7	ARCHITECTURE.md §10 gap 7 (N18)	"The recorded search gap understates the proven bound ... must be fixed before the composition root lands"	Fixed by Phase 9 F9-4. <code>unexploredRingDistance</code> is set to <code>ring</code> on <code>entr</code> <code>ring+1</code> only after a full ring; unvisited cells contribute their own floors and lower is taken (verified this pass at <code>expansion.js:465,477,691</code>). What si L3, one layer up
DD-8	ARCHITECTURE.md §10 gap 13	"the TLA+ configurations are not executed here"	All six were executed 2026-08-31. Two now close
DD-9	ARCHITECTURE.md §10 gap 9	"Inert today, live the moment a scheduler runs"	The scheduler runs. 5 of the 11 workers start today; two demonstrably used coded intervals
DD-10	gates.js <code>notEstablishedReason</code>	" <code>lifecycle.tla</code> has never been run under TLC"	False since 2026-08-31. Deliberately not corrected — the file is in the digi and the string understates rather than overstates the gap. Correct it at the when the digest moves anyway
DD-11	README.md "Current status"	"19 engine workers — none on production scheduling"; "185 modules"	12 of 19 start. Registered as cross-phase discrepancy C1; the module cou
DD-12	Phase 9 §9 L3 → Phase 10	"Recommend Phase 10 propagate rather than coerce"	Phase 10 closed without doing it and without mentioning it. Not a contra a dropped handoff, which is worse, because nothing tracks it

Historical record — must NOT be changed

- Every file in `docs/phase15/archive/` — preserved byte-for-byte, and several contain a later pass's correction of an earlier pass. That correction history is the evidence.
- The struck-through rows in the B-M results table and the "original entry, retained verbatim below" blocks for X6 and X7. They record what was believed and why.
- The 2026-08-15 TLC provenance discrepancy. Do not resolve it by editing either document.
- `PHASE_5_REMEDIATION_AND_CLOSURE.md` alongside `PHASE_5_ADVERSARIAL_REMEDIATION_AND_CLOSURE.md` — the first closure's own deferral sentence was the defect the second found.
- The two stale `phase5ExpirySemantics` FINDING assertions. V-9 deliberately refused to invert them; that refusal is itself the record.
- The `README.md` Phase 0 note — that no independent verification was ever written, and that writing one retrospectively would be a fabricated artefact.

15 V1 critical path

Four batches. Not twenty prompts.

BATCH	OBJECTIVE	FILES / SUBSYSTEMS	WHY V1	DEPENDS ON	VERIFICATION	CLAUDE ALO
B0 DO FIRST	Commit the formal work and fix L3. Commit <code>formal/lifecycle.tla</code> + <code>README.md</code> + the five canonical docs; then propagate the unproven search gap instead of coercing it	<code>formal/lifecycle.tla</code> · <code>src/engine/solve/round.js:248</code> · <code>tests/engine/solveRound.test.js</code>	Known correctness defect on the core path (L3), and the only verified formal evidence V1 has currently exists in an uncommitted file	Nothing	<code>lifecycle_c1</code> re-run as checked in · L3 mutant killed · engine lane green	YES — sta
B1	Owner decision gate. Record the V1 traversal source (OA-1), the V1 region (OA-2), and the capacity-1 boundedness acceptance (OA-3), each in an ADR or the freeze document	<code>docs/adr/ADR-35-*</code> (new) · a V1 freeze document	Without it the core path has no leaf and no shard. Nothing downstream can start	Nothing	The decision exists as a dated, recorded artefact	NO — owne
B2	Build the routing client and close the composition root. <code>routing/client.js</code> implementing §5.2's degradation ladder, §18.3 B6's uniform-treatment	<code>src/engine/routing/client.js</code> (new) · <code>server.js</code> · <code>src/workers/leaderWorkers.js</code> · <code>src/workers/coordinator.worker.js</code> · <code>src/engine/routing/chargerReachabilityCache.js</code>	The required core dependency is missing and the core path cannot	B1	<code>gate:composition</code> exit 0 — the single mechanical signal that V1's core works	YES, afte

BATCH	OBJECTIVE	FILES / SUBSYSTEMS	WHY V1	DEPENDS ON	VERIFICATION	CLAUDE ALO
	rule and the route(parts) seam; then construct evaluateExact , pricedCandidateFor , hopsForSequence at server.js and remove coordinator from UNCOMPOSABLE . Add the coordinator's cutover- switch guard (B1 item 5) and call assertVersionInKey (A9)		execute end to end. Discharges A9 and B1 item 5 as a consequence, not as separate work			
B3	Publish the V1 configuration and prove one round end to end. Publish a version binding the region, the shard and cutover.engine_enabled ; build a harness that issues a real POST /api/tasks/assign against a disposable cluster and follows it to a durable Commitment row	prisma config publish path · tools/verify/v1EndToEnd.js (new) · package.json	"Can a real request traverse it" is the V1 definition. Today the answer is no, and no harness has ever asked	B2	One request → Leg QUEUED → round claims → candidates → commit → dispatch, with a decision record carrying a proven gap	YES
B4	V1 freeze. Re-run the full verification set, re-measure the digest, and write one V1 freeze document that states what V1 is, what it is not, and every known limitation by name	npm test · npm run gates · 6 live-DB harnesses · V- 9 scope re-derivation · mutation pass · docs/V1_FREEZE.md (new) · optionally .github/workflows/ci.yml and gates.js 's stale string	A release whose limitations are not written down is not "with known limitations explicitly documented"	B3	VV-1 ... VV-8 all pass; the freeze document names the digest, the commit, and every V2 item	YES

IS THERE V1 IMPLEMENTATION WORK? YES — BUT ONLY ONE BATCH OF
IT, PLUS ONE THAT IS TRULY SMALL

B0 is small and available now. B2 and B3 are real engineering, and both are gated behind a single
owner decision. **There is no V1 work that requires an external vendor, a fleet, an attestation, a
calibration owner, or compute.**

16 Finite V1 stop condition

Once these 7 things are complete and these 6 checks pass, *stop*. Freeze V1.

THE 7 THINGS

1 The X7 formal work is committed

— formal/lifecycle.tla

, formal/README.md

and
the
five
canonical
documents
are
in
the
tree,
not
in a
working
copy.

2 L3 is fixed

— round.js

prop: achievedGapMilliCU: null

and achievedGapProven: false

, ?? 0n

and

a

muta

restor

is

killed.

3 The V1 traversal source is recorded in an ADR

(OA- *not*

1),

with

its

scope

and

its

limitations

stated,

and

with

an

explicit

sentence

that

it is

B1

Step

5.

4 The V1 operating region is declared

(OA-

2),

and

the

sub-

cell

coverage

consequence

is

stated

rather

than

worked

around.

5 The capacity-1 boundedness is accepted in writing

(OA- MaxTicks = 3

3)

—

as

a

global

budget

over

2

Legs.

6 `src/engine/routing/client.js` exists and the composition root injects it

```
, assertVersionInKey
includ
the
coord
cutov
switc
guard
and
the

call.
```

7 A V1 freeze document exists

naming
the
commit,
the
digest,
what
V1
is,
what
V1
is
not,
and
every
V2
item
by
ID
from
§3
of
this
audit.

THE 6 CHECKS

1 `npm run gates`

→ **exit 0**

```
, all gate:composition
8,

included.
```

2 `npm test`

→
exit
0, 0
failures,
0
skips.

3 One real

POSTreaches a durable
/api/tasks/assign

Commit to a disposable cluster, and its decision record carries a
proven search gap.

4 All six live-DB harnesses green, plus the new end-to-end one.

5 `lifecycle_c1`

and `commitment_c1`

re- *as checked in*

run

at

the

freeze

commit

—

both

close,

all

declared

properties

PASS.

6 Every mutant built for this work is killed, and the tree is byte-verified as restored.

After that point, the rule is absolute. A new observation becomes V2 automatically *unless* it demonstrably breaks the V1 core: a real request cannot traverse the path, or the path produces a wrong or unsafe result, and you can show it. Nothing else re-opens V1 — not a RED gate, not missing evidence, not an unsigned attestation, not an unconverged model configuration, not a stale document, not a module named in a plan and never built, and not the fact that a further pass would find something. *Six passes have each found defects on the surface the previous pass's fix created. That is structural, and it is the reason the stop condition is a count and not a judgement.*

17 What must not be touched for V1

DO NOT	WHY
Write a stub, fake or in-process router to make <code>gate:composition</code> pass	Starting the coordinator on invented travel times assigns real work on invented data. V1 recorded owner declaration, which is a different act
Select, rank, recommend or hint at a production routing engine	B1 Step 5 is a decision on recorded benchmark evidence, in an ADR. No evidence exists
Derive, edit, promote or reclassify any calibration value	§22.3 forbids any automated process from changing a Safety-class parameter
Weaken a gate, change the gate algebra, or convert <code>NOT_EVALUATED</code> → <code>GREEN</code>	<code>NOT_EVALUATED</code> blocks exactly as RED and is kept distinct on purpose
Remove the <code>[NOT PROVEN]</code> annotation or <code>establishedByCommand</code>	Only after all six exhaustive runs complete. Two have
Edit any <code>.cfg</code> , or add a boundedness, timeout or artificial-progress constant to a <code>.tla</code>	The configuration under check is part of the requirement (§24.2)
Modify <code>gates.js</code> , <code>evidence.js</code> , <code>sourceDigest.js</code> or any test to accommodate a TLC run	Both structural gaps are recorded deliberately and are not defects to close
Put <code>docs/</code> into the source-digest scope, or build a prose-parsing runbook gate	The first voids a 25-minute evidence collection on every prose edit; the second is an unmaintained future false green
Fabricate an observation window, soak record, fidelity study, shadow result, rehearsal record or attestation	P15-E1 exists because malformed windows were being ADMITTED and PASSED on exa gates
Invent a §4.2 TASK transition table, a mobility model, an extract vintage, a supply definition, or the §17.4 "affected classes" set	The specification is FROZEN. Each is a decision, and a producer that manufactures its c is the failure this project records rather than closes
Treat <code>SEED_SPATIAL_MAP</code> / <code>RGN-BLR</code> as production configuration	It is a Phase 2 containment demonstration — four fine cells, placeholder ids, never pub config version
Fabricate the missing <code>Mission</code> row to make <code>phase12LiveDatabase</code> green	The seed defect is a §22.1 governance choice, not a fixture to paper over
Edit the two stale <code>phase5ExpirySemantics</code> assertions to their inverse	V-9 refused to, deliberately. Phase 5 owns its own harness
Enable any Tier 2 kill switch, or begin Phase 16	§1.8 rule 3. Phase 16's prerequisite is Phase 15, and V1 is not Phase 15
Re-open a closed finding because an archived report shows it OPEN, or close one because an archived report says FIXED	The archive is frozen history

18 Final recommendation

The decision table

ITEM	V1 / V2	WHY	CLAUDE?	BLOCKS CORE RUN
L3 – round.js:248	V1 A	Known correctness defect on the core path; caller already exists	Yes	Corrupts every de record
Commit X7 formal work	V1 B	V1's only formal evidence is in an uncommitted file	Yes	No, but it is unrec
V1 traversal source (OA-1)	V1 A	The core path has no leaf	No	Yes — totally
V1 region (OA-2)	V1 A	No shard to enable; the gate refuses every request	No	Yes
Boundedness acceptance (OA-3)	V1 B	§7.3a item 8 must be signed, not assumed	No	No
routing/client.js + composition root	V1 A	Required core dependency missing	Yes, after OA-1	Yes
A9 · B1 item 5	V1 A	Both discharge as consequences of the above, not as separate work	Yes, with it	Latent until live
End-to-end harness	V1 B	"Can a real request traverse it" is the V1 definition and has never been asked	Yes	It is the proof
B1 Steps 1/3/4/5 · D3 · D8	V2 E	Procurement and production quality, not runtime existence	No	No
B8 – 39 findings	V2 E	Calibration incomplete with a working runtime is excluded by your own rule	No	No
B-P · B-0 · V-5 · OP-8 · F4	V2 D	Observation, attestation, evidence currency	No	No
lifecycle_c2/c3 · commitment_c2/c3	V2 G	Capacity 2/3 checks chaining — Tier 2, switch thrown, Phase 16e's gate	Mechanically	No
X3 · OD-2/3/4/5 · §20.5 · D1-esc · seed governance	V2 F	Frozen specification; a decision, not a commit	Must not	No
N16 · N17 · stores/roles.js · deps/*	V2 C	Real, open, and none breaks correctness of the shipped path	Yes	No
Phase 16a-16i · all Tier 2	V2 G	Nine modules and §16 reliability must be built inside it	Not now	No
Seed · N13 · P15-F7a · CI · doc drift	RESIDUAL I	None permissive; none on the decision path	Mostly yes	No

A · THE SMALLEST REALISTIC DEFINITION OF V1

A working core assignment engine that executes its Tier 0 + Tier 1 workflow end to end at capacity 1, against declared operating region and one declared traversal source, with its limitations written down.

Not a certified production release. Not "every §24 gate green" — 8 of those gates are discharged by observation, attestation, calibration or procurement, and none of them is a statement about whether the engine works. Capacity 1 is the correct scope because chaining is Tier 2 with its kill switch thrown.

B · WHAT ACTUALLY REMAINS TO MAKE THAT V1 WORK

Four things. (1) Decide the traversal source. (2) Declare the region. (3) Build routing/client.js and close the composition root so gate:composition exits 0. (4) Fix L3 so the decision record stops advertising an optimality it did not prove. That is the whole list. Everything else in this repository is either already done, or is V2.

C · WHAT CLAUDE CAN COMPLETE

The L3 fix and the formal commit today, with no dependency. Then, once you have made one decision: the routing client, the composition root, the coordinator's cutover guard, the assertVersionInKey call, the configuration publish, the end-to-end harness, all verification, and

the freeze document. **Roughly 90% of the remaining V1 effort is mine, and 100% of it is gated behind one sentence from you.**

D · WHAT REQUIRES YOU OR THE REAL WORLD

For V1: three things, all of them writing. OA-1, OA-2, OA-3 — the traversal source, the region, the boundedness acceptance. **For V2: everything the repository has correctly refused to fake** — a benchmarked engine selection, a fleet, a calibration owner, three attestations, a rehearsal, compute for two model configurations, and a set of frozen-specification decisions.

E · WHAT SHOULD EXPLICITLY MOVE TO V2

All 8 non-green §24 gates. All of B8, B-P, B-O. B1 Steps 1/3/4/5, D3, D8. `lifecycle_c2/c3` and `commitment_c2/c3`. X3 and every OD. §20.5's four semantics. N16, N17, `stores/roles.js`, `deps/*`. Every Tier 2 mechanism and all of Phase 16. Every residual observation. **Write them into the V1 freeze document by ID so the boundary is a record and not a memory.**

F · WHAT MUST NEVER BE ATTEMPTED TO LOOK COMPLETE

A stub router. A fabricated observation window, soak, fidelity study, shadow result, rehearsal or attestation. A derived Safety-class value. A weakened gate or a `NOT_EVALUATED` flipped green. A `.cfg` edited or a bound added to make TLC converge. A manufactured caller for a guard whose seam does not exist. An invented region, speed model, extract vintage, supply definition or "affected classes" set. A retrospective Phase 0 verification. **Each of these has a specific past defect behind it, and each would make the project look finished while making it less true.**

G · THE V2 ROADMAP, AFTER THE V1 FREEZE

V2-a · Unblock the externals in parallel — name a calibration owner (B8), resolve the D1 coverage escalation, answer D3 and D8. **V2-b · Run B1 properly** — Steps 1/3/4, then the Step 5 ADR, then swap the V1 traversal source for the selected engine behind the same seam. **V2-c · Start the observation clocks** — shadow is 14 days and cannot begin until the composition root lands, which V1 delivers; then soak, invariants, fidelity. **V2-d · Close the formal remainder** on provisioned compute. **V2-e · File the three attestations and rehearse the rollback.** **V2-f · Close Phase 15, then scope Phase 16 as the build phase it actually is.** Fold N16, N17, `stores/roles.js` and `deps/*` into V2-b, where the round is finally running and their benefit is measurable.

H · WHAT V1 IS BLOCKED BY, PRECISELY

A combination of exactly two things, and neither is verification.

One external decision — what `route(parts)` talks to and which region it serves — and **the code that decision unblocks**: one module, one composition-root wiring, one line in `round.js`.

V1 is not blocked by verification. The suite is green, the gates are green but for the one that measures this exact gap, the live-DB harnesses are 167/167, cross-phase re-verification found no regression, and both capacity-1 formal configurations close exhaustively with every declared property passing. **V1 is not blocked by specification** — X3 and the ODs touch nothing on the core path. **V1 is not blocked by an external dependency in the procurement sense** — it needs a declared source, not a purchased one. It is blocked by one decision you are entitled to make, and about two days of work behind it.

Audit method. Documentation read first, then verified against the repository by execution. Re-measured live this pass: source digest (`d033038c...` /573), worker registry (19/9/4), `gate:composition`, `release:verdict` (0/17/7, every RED [STALE]), `routing:readiness`, `gate:calibration` (250/52/160/38, 54 Safety-class, 39 findings), `sim:fidelity` (7

NOT_MEASURED), defaultSnapshot() (spatial null / shards null / 0 bindings), the cadence-parameter intersection (11 of 19 workers), the empty stores/ and deps/ directories, the seed changeClass defect, expansion.js 's bound-then-exact call and its ring arithmetic, round.js:248 , and git status /HEAD.

Not re-run in this read-only pass (cited from the canonical record, dated): npm test 162/7 275, the six live-DB harnesses 167/167, V-9's 315/317, and every TLC state count. safety:case was deliberately not run because it regenerates a file in place.

TRUTH > GREEN. No conflict in this report was silently reconciled; where two documents disagree, both are shown.